

Active Directory Domain Service, DNS and DHCP Configuration

After completing the initial server configuration, the Windows Server 2022 (DC01) was configured with Active Directory Domain Services (AD DS), DNS, and DHCP to build the core infrastructure for the enterprise lab environment.

1. Local Server Preparation

Before installing any roles, the server was prepared using GUI-based configuration.

Configuration Procedure

1. Open **Server Manager**.
2. Go to **Local Server**.
3. Configure the following settings:
 - Computer Name
 - Time Zone
 - Desktop Wallpaper
 - Static IP Address

Static IP Configuration

The server was configured with a static IP address for domain controller stability.

Parameter	Value
IP Address	192.168.1.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1 (pfSense LAN Gateway)
DNS Server	192.168.1.10

Result

The server was successfully configured with hostname, timezone, wallpaper, and static IP address.

2. Active Directory Domain Services (AD DS)

Active Directory Domain Services were installed to create a centralized authentication system.

The server was promoted to Domain Controller under a new forest.

Configuration Procedure

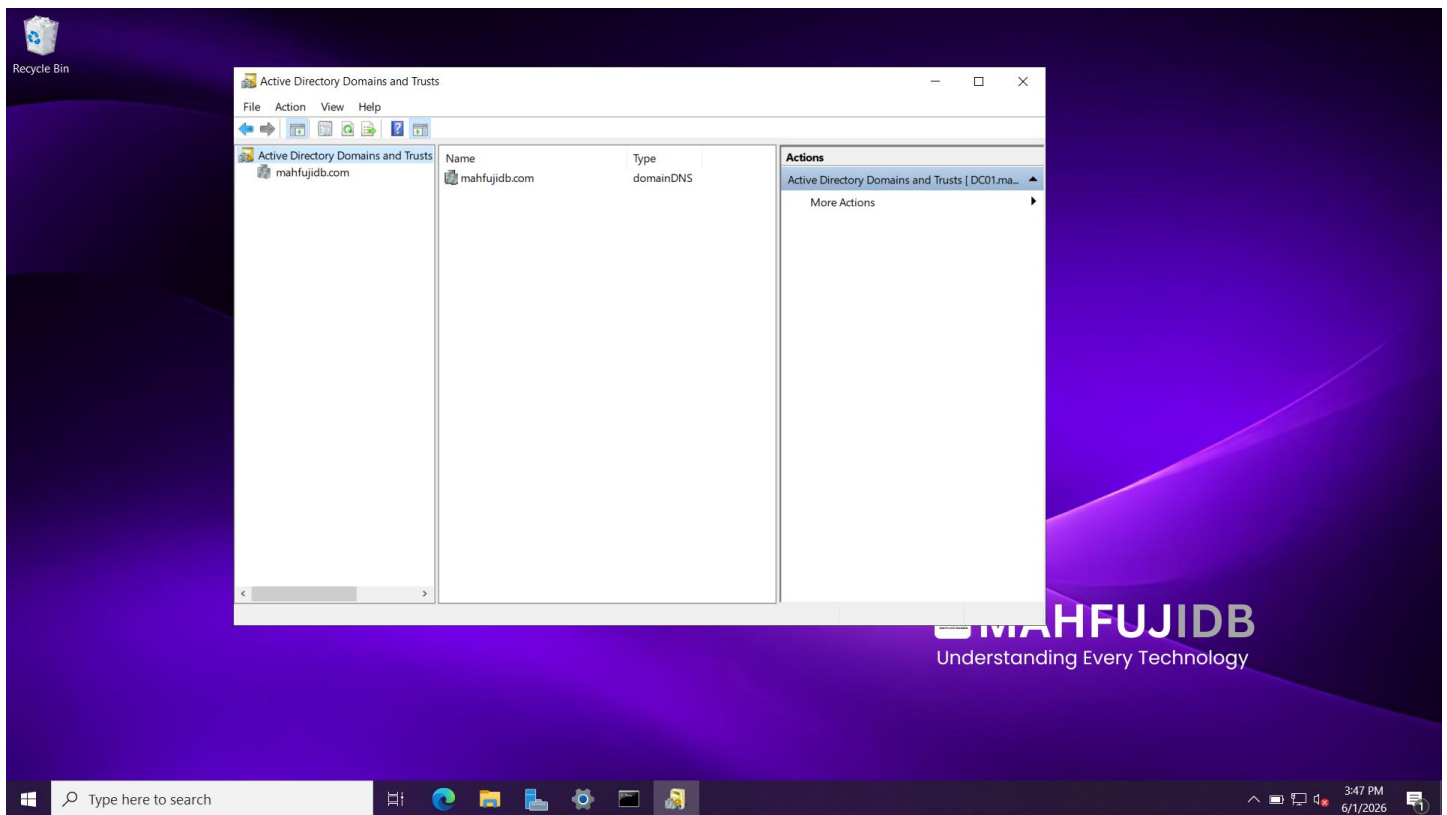
1. Open **Server Manager**.
2. Select **Add Roles and Features**.
3. Choose **Role-based installation**.
4. Select server (**DC01**).
5. Select **Active Directory Domain Services**.
6. Add required features.
7. Install the role.
8. Click **Promote this server to a domain controller**.
9. Select **Add a new forest**.
10. Enter domain name:

mahfujidb.com

11. Set DSRM password.
12. Complete installation and restart server.

Result

Active Directory Domain Services were successfully installed and the server was promoted as a Domain Controller for **mahfujidb.com**.



3. DNS Server Configuration

DNS was installed with AD DS and configured for domain name resolution.

Configuration Procedure

1. Open **Server Manager** → **Tools** → **DNS**.
2. Verify Forward Lookup Zone:

mahfujidb.com

3. Confirm DNS server IP:

192.168.1.10

4. Ensure DNS is integrated with AD DS.

Reverse Lookup Zone Configuration

Five reverse lookup zones were created for departmental networks.

Reverse Lookup Zones

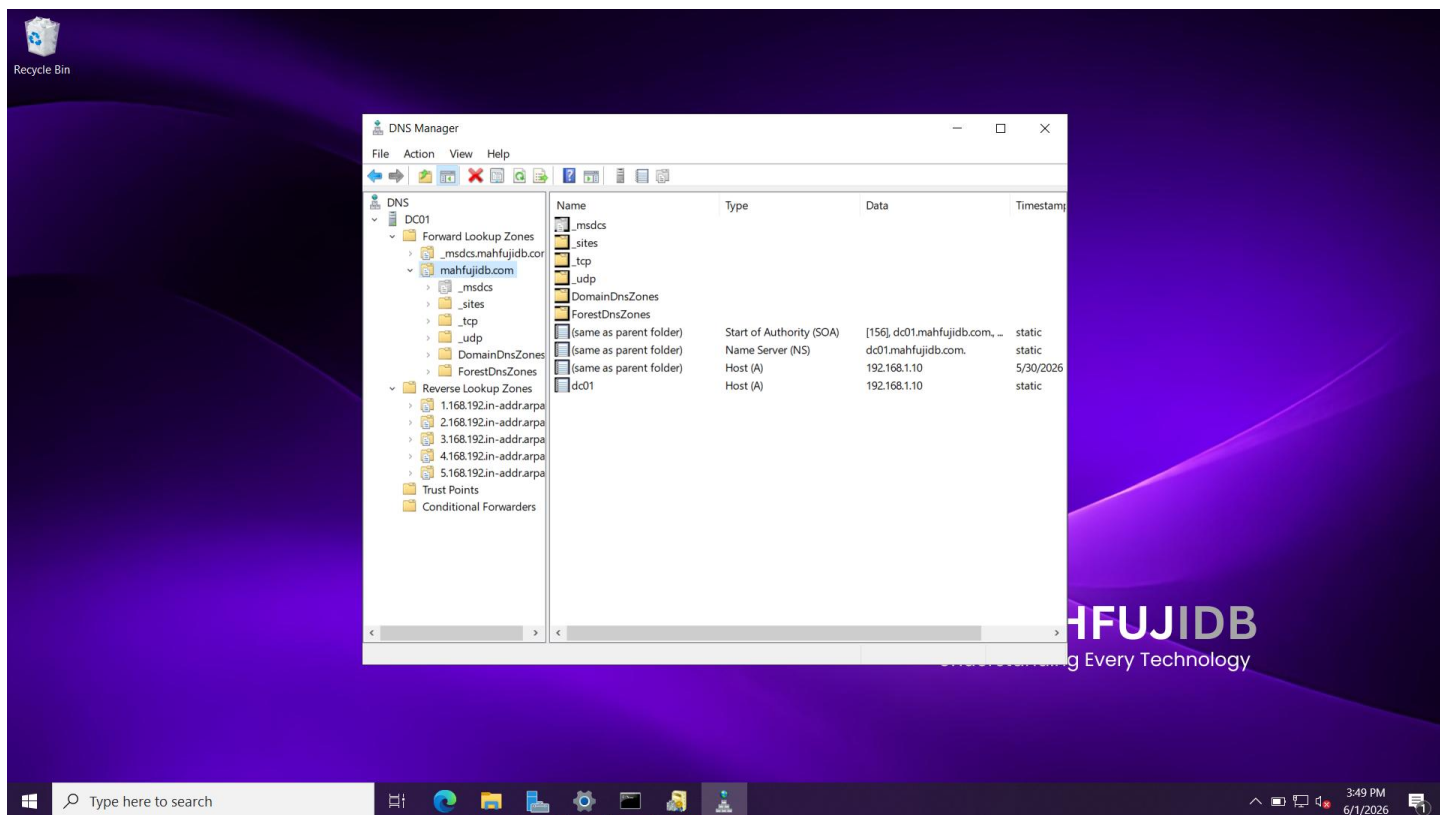
Zone Name	Network
1.168.192.in-addr.arpa	SERVER Network
2.168.192.in-addr.arpa	MGMT Network
3.168.192.in-addr.arpa	FIN Network
4.168.192.in-addr.arpa	HR Network
5.168.192.in-addr.arpa	IT Network

Configuration Procedure

1. Open **DNS Manager**.
2. Expand server (**DC01**).
3. Right-click **Reverse Lookup Zones**.
4. Click **New Zone**.
5. Select **Primary Zone**.
6. Select **IPv4 Reverse Lookup Zone**.
7. Enter network ID for each department.
8. Repeat for all five networks.

Result

Five reverse lookup zones were successfully created for reverse DNS resolution across all departmental networks.



4. DHCP Server Configuration

DHCP was installed and configured to automatically assign IP addresses for all departments.

DHCP Scope Configuration

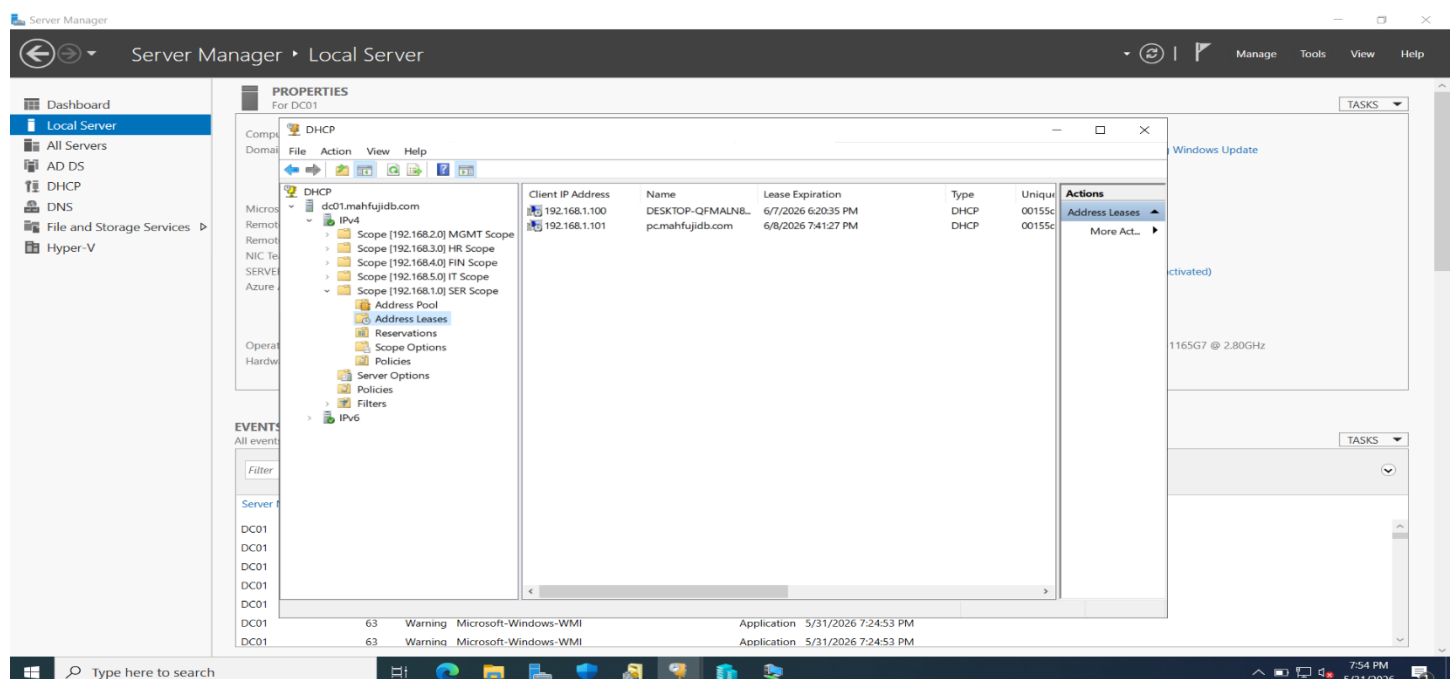
Scope Name	IP Range
SERVER-SCOPE	192.168.1.100 – 192.168.1.200
MGMT-SCOPE	192.168.2.100 – 192.168.2.200
FIN-SCOPE	192.168.3.100 – 192.168.3.200
HR-SCOPE	192.168.4.100 – 192.168.4.200
IT-SCOPE	192.168.5.100 – 192.168.5.200

Configuration Procedure

1. Open **Server Manager**.
2. Add **DHCP Server Role**.
3. Complete installation.
4. Open **DHCP Console**.
5. Create IPv4 scopes for each department.
6. Configure subnet mask and IP ranges.
7. Activate all scopes.
8. Authorize DHCP in Active Directory.

Result

DHCP server was successfully configured and activated for all departmental networks.



5. pfSense DHCP Relay Configuration

pfSense was configured as a DHCP relay agent to forward DHCP requests to the centralized DHCP server.

Configuration Procedure

1. Open **pfSense Web Interface**.
2. Navigate to **Services** → **DHCP Relay**.
3. Enable DHCP Relay.
4. Select interfaces:

- OPT1 (MGMT)
- OPT2 (FIN)
- OPT3 (HR)
- OPT4 (IT)

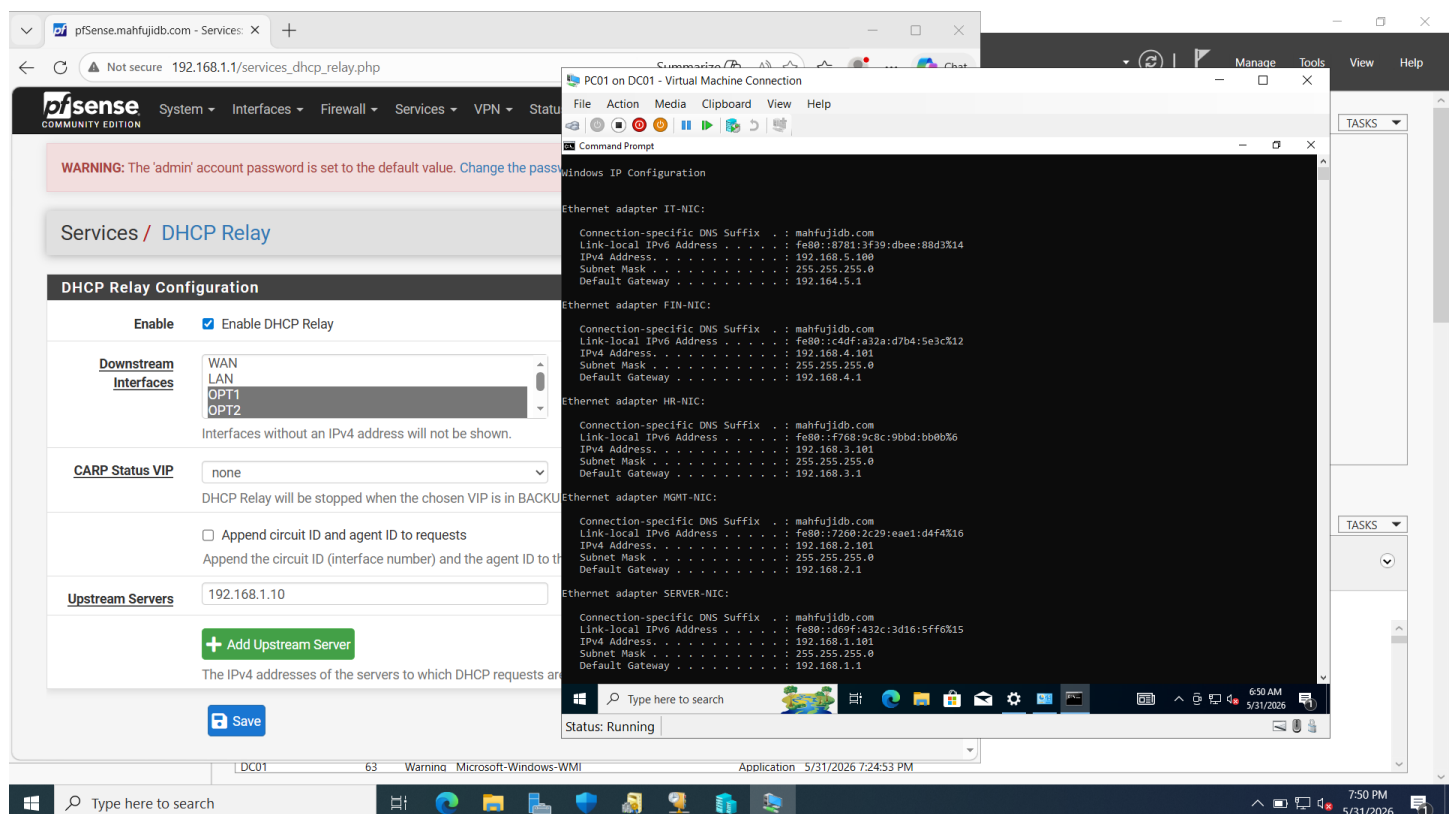
5. Set DHCP Upstream server IP:

192.168.1.10

6. Save and apply configuration.

Result

pfSense successfully forwarded DHCP requests from all departmental networks to the centralized DHCP server.



6. Client Machine DHCP Verification

Client virtual machines were created to verify DHCP, DNS, and domain network functionality.

Client Configuration

Client	Network Switch
SERVER-PC	SERVER-SWITCH
MGMT-PC	MAMT-SWITCH
FIN-PC	FIN-SWITCH
HR-PC	HR-SWITCH
IT-PC	IT-SWITCH

Testing Procedure

1. Start client VM.
2. Set NIC to **Obtain IP automatically**.
3. Run command:

```
ipconfig /all
```

Result

All client machines successfully received:

- IP Address from DHCP
- Default Gateway from pfSense
- DNS Server (192.168.1.10) from AD DS
- Domain connectivity to **mahfujidb.com**

Final Result

The enterprise infrastructure successfully achieved:

- Windows Server 2022 Domain Controller setup
- Active Directory Domain Services (mahfujidb.com)
- DNS with forward and reverse lookup zones
- 5 reverse DNS zones configured
- Centralized DHCP service
- pfSense DHCP relay configuration
- Fully automated IP assignment across departments
- Complete network segmentation with pfSense firewall